

Project Proposal Form, Fall 2026
Institute of Information Technology (IIT)
University of Dhaka

Student's Name:	Md. Habibur Rahman		
Student's Roll:	220	Phone:	[Enter current phone number]

Project Title:

Design, Development and Evaluation of a Specification-Driven and Reusable Alumni Management Platform for a Resource-Constrained Association

Motivation:

Alumni associations often depend on paper records, spreadsheets, informal messaging groups and manually maintained financial records. These practices make membership verification, payment tracking, communication, event coordination and governance-document management difficult.

The Govt. Haraganga College Alumni Association needs a low-cost, maintainable and privacy-aware system of record that volunteers can operate. The platform must work without enterprise-level staffing, recurring software licences or live payment-gateway credentials.

This project focuses on reliable membership records, financial transparency, privacy-aware directory access, event and communication management, governance documents, mobile-first access, single-maintainer sustainability, specification-driven development and measurable testing. It also examines how far Association rules can be represented in software without allowing software to replace the authority of the constitution or Election Commission.

This is not only a CRUD application. Its research and engineering focus is the combination of constitutional rules represented as traceable and testable business rules, manual evidence-based payment as a deliberate primary workflow, a reusable and configurable platform foundation, operation by one maintainer at a low annual cost, portability across supported database providers and container hosts, safe handling of legacy databases and archived records, and quality evaluation against functional and non-functional requirements.

Project Description:

GHCAA is a multi-client platform consisting of an ASP.NET Core Web API (Clean Architecture), an Angular web application and a Flutter mobile application, with PostgreSQL for deployment and SQLite for local development and tests.

The project follows specification-driven development. Requirements are derived from stakeholder needs, current manual processes, Association documents and software-engineering standards, and each important requirement is traced to its source, business rule, implementation component, automated test and evaluation result.

The main functional areas are: membership and identity (registration, approval, digital ID, certificates, profiles, privacy controls, searchable directory); accounting and payments (fees, dues, donations, configurable payment instructions, manual payment proof, administrative verification, receipts and audit history); events and participation (publication, registration, capacity control, waitlisting, QR attendance, budgets and moderated photo submissions); communication and networking (news, notices, targeted communication, real-time messaging, notifications, jobs and mentorship); and governance and administration (constitution versioning, Executive Committee records, amendment voting, polls, voter-roll preparation and election-document support). The platform supports governance procedures but does not replace the Election Commission or claim to provide a legally sufficient electronic election system.

The platform is built as a reusable foundation rather than a one-off application. Authentication, role-based authorisation, accounting and ledger services, audit logging, file handling, SignalR notifications, organisation configuration, shared validation, reusable Angular controls and Flutter widgets, and CI/CD are designed as organisation-agnostic services, separate from Association-specific rules and configuration. Runtime settings such as branding, currency, feature flags and payment instructions can be changed by administrators without redeployment, and a boot profile pack can seed a fresh deployment's defaults, documents and content.

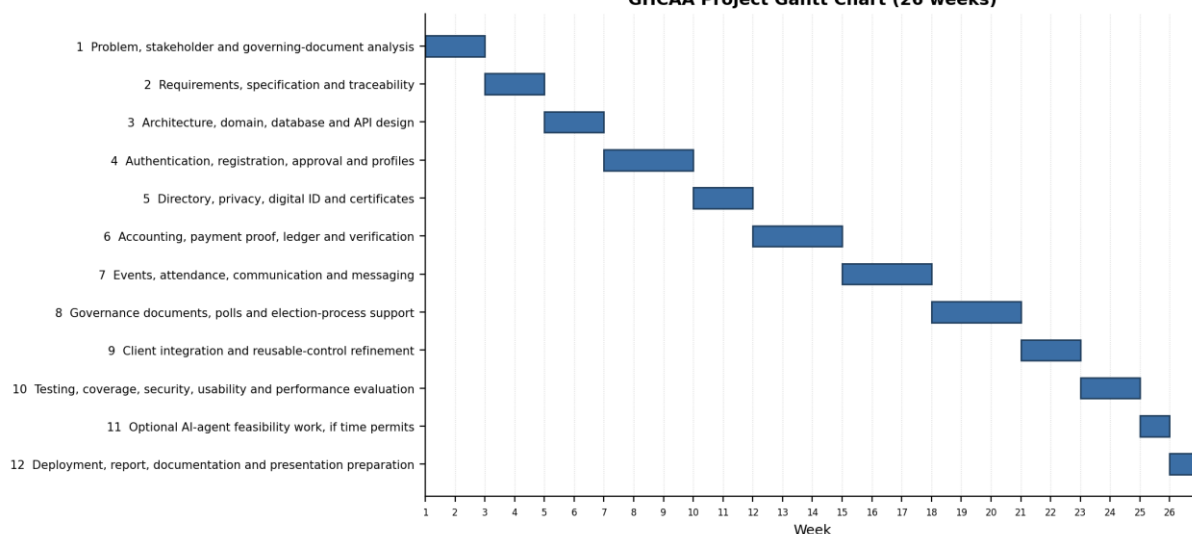
Non-functional requirements cover performance, security, reliability, usability, maintainability, portability, recovery, auditability and privacy, evaluated through quality-attribute scenarios such as directory latency, session invalidation on status change, dependency-failure isolation, architecture-rule enforcement and recovery time. The prototype already includes the backend, web app, mobile app, authentication, accounting, membership, events, governance and CI/CD. Remaining work covers traceability closure, measured coverage, security and usability evidence, performance evaluation and documentation.

Strategy & Timeline:

The project Gantt chart is given below:

Week	Task No / Module	Description	Dependencies
1	WP1 – Problem & stakeholder analysis	Stakeholder needs, current manual processes and governing documents (constitution, election rules) reviewed and analysed.	—
2	WP1 – Problem & stakeholder analysis	Governing-document analysis completed; problem statement and scope finalised.	Week 1
3	WP2 – Requirements & specification	Requirements derived from stakeholder needs and standards; traceability approach defined.	Week 1-2
4	WP2 – Requirements & specification	Requirement traceability (source -> business rule -> component -> test) drafted for core modules.	Week 3

GHCAA Project Gantt Chart (26 weeks)



Languages or Tools to be used:

- Frontend: Angular, TypeScript, HTML, SCSS, Angular Signals
- Backend: ASP.NET Core 9 Web API, C#, Clean Architecture, PostgreSQL (deployment) / SQLite (local & tests)
- Database: PostgreSQL (deployment), SQLite (local development and tests)
- IDE / Editor: Visual Studio Code
- AI Tools / Assistants (if any) — e.g., GitHub Copilot, ChatGPT, Claude: GitHub Copilot (via Copilot SDK in VS Code), Claude, ChatGPT

AI Tool Use Declaration:

☒ I intend to use AI coding/writing assistance tools for this project.

☐ I do not intend to use AI tools for this project.

If yes, tool(s) to be used: GitHub Copilot (Copilot SDK, VS Code), Claude, ChatGPT.

Intended purpose(s) (e.g., boilerplate generation, debugging, documentation drafting, learning a new framework): Framework/library learning, boilerplate generation, debugging, test suggestions, code review, documentation drafting, technical comparison and explaining implementation decisions. All AI-assisted output is reviewed, tested and modified by the student, who remains responsible for requirements, architecture, implementation, testing and security decisions.

Declaration: I understand that, regardless of AI tool use, I am required to disclose such use in my project report, and I must be able to independently explain, defend, and modify any part of my code — including any AI-assisted sections — during the midterm and final presentations, live defense, and supervisor meetings throughout the semester. I understand that submitting AI-generated work I cannot explain or defend may result in a reduced grade or other academic consequences.

Student's Signature: _____

Date: _____

Project GitHub link: <https://github.com/shalin-rahman/GHCA>

Note: Maintain a regular, incremental, and dated commit history throughout the semester. A single large commit submitted close to the deadline is treated as a red flag and may be investigated.

Supervisor's Name: _____

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for the midterm presentation.

I further confirm that the student has demonstrated independent understanding of the codebase — including any AI-assisted portions — and was able to explain and modify their prototype during our supervision meetings.

Signature of the supervisor: _____

Date: _____

Before the Final Presentation:

I confirm that the progress is satisfactory and I am forwarding it for the final presentation.

I further confirm that the student has consistently demonstrated understanding and ownership of their work — including any AI-assisted code — throughout the semester, based on regular supervision meetings and progress reviews.

Signature of the supervisor: _____

Date: _____

Proposal Presentation Feedback (if any):

Midterm Presentation Feedback (if any):

AI-Use Disclosure Review Notes (if any):

(For supervisor/evaluator use — note any discrepancy between disclosed AI use and demonstrated understanding.)